Final exam · Unit 6 · Laplace equation by separation · 2.5 points · 50 min · Standard exam level

P21 — Laplace's equation in a circular wedge with one edge held at zero and the other insulated: angular eigenfunctions, bounded radial powers and a datum that is a hidden finite sum

A thin plate has the shape of the circular sector $D = \{(r,\theta) :\ 0 < r < a,\ 0 < \theta < \alpha\}$*, with* $0 < \alpha \le 2\pi$*. Its faces are insulated, so its steady temperature* $u(r,\theta)$ *solves*

$$\Delta u = u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} = 0 \quad \text{in } D,$$

$$u(r,0) = 0, \qquad u_\theta(r,\alpha) = 0 \quad (0 < r < a), \qquad |u(r,\theta)| \text{ bounded as } r \to 0^+, \qquad u(a,\theta) = f(\theta) \quad (0 < \theta < \alpha).$$

The straight edge $\theta = 0$ *is held at temperature* 0*, the straight edge* $\theta = \alpha$ *is insulated, and the arc* $r = a$ *has a prescribed temperature.*

(a) Separate variables, $u = R(r)\Theta(\theta)$*, and solve the angular eigenvalue problem, discussing the three possible signs of the separation constant. Show that the eigenfunctions are* $\Theta_n = \sin\left(\nu_n\theta\right)$ *with* $\nu_n = \dfrac{(2n-1)\pi}{2\alpha}$*,* $n = 1, 2, \dots$

(b) Solve the radial equation, justify which solutions are admissible, and write the general bounded solution together with the formula for its coefficients in terms of f*.*

(c) Let $\alpha = \pi/2$ *(a quarter disc) and* $f(\theta) = \sin^3\theta$*. Find* $u(r,\theta)$ *in closed form, check all the conditions, write it in Cartesian coordinates* $x = r\cos\theta$*,* $y = r\sin\theta$*, and give* u *at* $(r,\theta) = (a/2, \pi/2)$ *and at* $(a/2, \pi/6)$*.*

Setup and canonical formulation

Phase 1. Hypotheses and setup

- Unknown $u(r,\theta)$, independent variables $r \in (0,a)$, $\theta \in (0,\alpha)$; $u \in C^2(D)$, continuous up to the boundary except possibly at corners.
- Linear, homogeneous, second-order elliptic PDE (Laplace). Homogeneous conditions on the two straight edges (Dirichlet at $\theta = 0$, Neumann at $\theta = \alpha$) and a boundedness condition at the vertex $r = 0$; the only nonhomogeneous condition is on the arc $r = a$. This is the structure that allows separation of variables with eigenfunctions in θ.
- Data: $a > 0$ (radius), α (opening angle), f (arc datum). In (c): $\alpha = \pi/2$, $f = \sin^3\theta$; u is a dimensionless temperature.

Phase 2. Canonical formulation

Multiplying the PDE by r^2: $r^2 u_{rr} + r u_r + u_{\theta\theta} = 0$. With $u = R(r)\Theta(\theta)$:

$$\frac{r^2 R'' + r R'}{R} = -\frac{\Theta''}{\Theta} = \lambda \quad (\text{constant}),$$

$$\Theta'' + \lambda\Theta = 0,\ \ \Theta(0) = 0,\ \ \Theta'(\alpha) = 0; \qquad r^2 R'' + r R' - \lambda R = 0,\ \ |R(0^+)| < \infty.$$

The angular problem is a regular Sturm-Liouville problem with $p = \sigma = 1, q = 0$ (Haberman Sec. 5.3).

Part (a) · 1 point

Separation and the angular eigenvalue problem

Step 1. Separation

$u_{rr} = R''\Theta, u_r = R'\Theta, u_{\theta\theta} = R\Theta''$. Then $r^2 R''\Theta + r R'\Theta + R\Theta'' = 0$; dividing by $R\Theta$ (where it is not zero),

$$\frac{r^2 R'' + r R'}{R} = -\frac{\Theta''}{\Theta}.$$

The left side depends only on r and the right only on θ, so both equal a constant λ. The edge conditions become $R(r)\Theta(0) = 0$ and $R(r)\Theta'(\alpha) = 0$ for all r; for a nontrivial R, $\Theta(0) = 0$ and $\Theta'(\alpha) = 0$.

Step 2. Case $\lambda = -\mu^2 < 0$

$\Theta = A\cosh\mu\theta + B\sinh\mu\theta$. $\Theta(0) = A = 0$. $\Theta'(\alpha) = B\mu\cosh\mu\alpha = 0$; since $\mu > 0$ and $\cosh\mu\alpha \geq 1$, $B = 0$. Trivial.

Step 3. Case $\lambda = 0$

$\Theta'' = 0, \Theta = A\theta + B$. $\Theta(0) = B = 0, \Theta'(\alpha) = A = 0$. Trivial: unlike the full disc, the constant is not allowed (it would violate $u = 0$ on $\theta = 0$).

Step 4. Case $\lambda = \nu^2 > 0$

$\Theta = A\cos\nu\theta + B\sin\nu\theta$. $\Theta(0) = A = 0$. $\Theta'(\alpha) = B\nu\cos\nu\alpha = 0$ with $B \neq 0, \nu > 0$, so $\cos\nu\alpha = 0$:

$$\nu\alpha = \frac{(2n-1)\pi}{2} \implies \nu_n = \frac{(2n-1)\pi}{2\alpha}, \qquad \Theta_n(\theta) = \sin\left(\frac{(2n-1)\pi\theta}{2\alpha}\right), \qquad n = 1, 2, \dots$$

Each Θ_n vanishes at $\theta = 0$ and has zero slope at $\theta = \alpha$: a quarter wave, three quarter waves, and so on across the opening.

Step 5. Orthogonality and norm

For $m \neq n$ the eigenvalues differ, so $\int_0^\alpha \Theta_m\Theta_n\, d\theta = 0$ (Haberman Sec. 5.3; the boundary terms $[\Theta_m\Theta_n' - \Theta_n\Theta_m']_0^\alpha$ vanish because $\Theta(0) = 0$ and $\Theta'(\alpha) = 0$). For $m = n$, with $\sin^2 s = \frac{1}{2}(1 - \cos 2s)$:

$$\int_0^\alpha \sin^2\nu_n\theta\, d\theta = \left[\frac{\theta}{2} - \frac{\sin 2\nu_n\theta}{4\nu_n}\right]_0^\alpha = \frac{\alpha}{2} - \frac{\sin\big((2n-1)\pi\big)}{4\nu_n} = \frac{\alpha}{2}.$$

KEY RESULTS

- $\lambda < 0$ and $\lambda = 0$: only $\Theta \equiv 0$.
- $\lambda = \nu^2 > 0$: $\Theta = \sin\nu\theta$ with $\cos\nu\alpha = 0$, so $\nu_n = \dfrac{(2n-1)\pi}{2\alpha}$, $\lambda_n = \nu_n^2$.
- $\displaystyle\int_0^\alpha \sin\nu_m\theta\, \sin\nu_n\theta\, d\theta = \frac{\alpha}{2}\delta_{mn}$.

Part (b) · 0.6 points

Radial equation and the general bounded solution

Step 1. Euler equation

With $R = r^p$: $R' = pr^{p-1}$, $R'' = p(p-1)r^{p-2}$, so $r^2R'' + rR' - \nu^2 R = \big[p(p-1) + p - \nu^2\big]r^p = (p^2 - \nu^2)r^p$. Hence $p = \pm\nu_n$ and, since $\nu_n \neq 0$, the two roots are distinct (Haberman Sec. 2.5):

$$R_n(r) = c_1 r^{\nu_n} + c_2 r^{-\nu_n}.$$

Step 2. Boundedness at the vertex

$\nu_n > 0$, so $r^{-\nu_n} \to \infty$ as $r \to 0^+$ while $r^{\nu_n} \to 0$. The condition $|u| < \infty$ at $r = 0$ forces $c_2 = 0$. (The $\lambda = 0$ branch $c_1 + c_2 \ln r$ does not appear, because $\lambda = 0$ is not an eigenvalue.)

Step 3. Superposition and coefficients

By linearity (Haberman Sec. 2.2), the sum of the product solutions is a solution of the PDE that satisfies the homogeneous conditions. Writing the constants as A_n / a^{ν_n} for convenience:

$$u(r, \theta) = \sum_{n=1}^{\infty} A_n \left(\frac{r}{a} \right)^{\nu_n} \sin \nu_n \theta.$$

At $r = a$: $f(\theta) = \sum_n A_n \sin \nu_n \theta$. Multiplying by $\sin \nu_m \theta$, integrating over $(0, \alpha)$ and using (a), Step 5:

$$A_m = \frac{\int_0^\alpha f(\theta) \sin \nu_m \theta \, d\theta}{\alpha / 2} = \frac{2}{\alpha} \int_0^\alpha f(\theta) \sin \nu_m \theta \, d\theta.$$

KEY RESULTS

- $R = r^p$ gives $p^2 - \nu_n^2 = 0$, $R = c_1 r^{\nu_n} + c_2 r^{-\nu_n}$; boundedness forces $c_2 = 0$.
- $u(r, \theta) = \sum_{n=1}^{\infty} A_n \left(\frac{r}{a} \right)^{\nu_n} \sin \nu_n \theta$, $A_n = \frac{2}{\alpha} \int_0^\alpha f(\theta) \sin \nu_n \theta \, d\theta$.

Part (c) · 0.9 points

Quarter disc with $f = \sin^3 \theta$

Step 1. Eigenvalues for the quarter disc

$\nu_n = \dfrac{(2n-1)\pi}{2 \cdot \pi/2} = 2n - 1$: eigenfunctions $\sin \theta, \sin 3\theta, \sin 5\theta, \ldots$, and radial factors $r, r^3, r^5, \ldots$

Step 2. The datum as a finite sum

$\sin 3\theta = \sin(2\theta + \theta) = \sin 2\theta \cos \theta + \cos 2\theta \sin \theta = 2 \sin \theta \cos^2 \theta + (1 - 2 \sin^2 \theta) \sin \theta$. With $\cos^2 \theta = 1 - \sin^2 \theta$: $\sin 3\theta = 2 \sin \theta - 2 \sin^3 \theta + \sin \theta - 2 \sin^3 \theta = 3 \sin \theta - 4 \sin^3 \theta$. Solving for the cube:

$$\sin^3 \theta = \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta.$$

Both terms are eigenfunctions, so by uniqueness of the coefficients (orthogonality) $A_1 = \frac{3}{4}$, $A_2 = -\frac{1}{4}$ and $A_n = 0$ for $n \geq 3$. Cross-check of A_1 with the formula of (b): $A_1 = \frac{4}{\pi}\int_0^{\pi/2} \sin^3\theta \sin\theta \, d\theta = \frac{4}{\pi} \cdot \frac{3\pi}{16} = \frac{3}{4}$, using $\int_0^{\pi/2} \sin^4\theta \, d\theta = \frac{3\pi}{16}$ (from $\sin^4 = \frac{3}{8} - \frac{1}{2}\cos 2\theta + \frac{1}{8}\cos 4\theta$).

Step 3. Solution

$$u(r,\theta) = \frac{3}{4}\frac{r}{a}\sin\theta - \frac{1}{4}\left(\frac{r}{a}\right)^3 \sin 3\theta.$$

Checks: $u(r,0) = 0$ because $\sin 0 = 0$; $u_\theta = \frac{3}{4}\frac{r}{a}\cos\theta - \frac{3}{4}\left(\frac{r}{a}\right)^3 \cos 3\theta$ vanishes at $\theta = \pi/2$ because $\cos\frac{\pi}{2} = \cos\frac{3\pi}{2} = 0$; u is bounded (a polynomial in r); $u(a,\theta) = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta = \sin^3\theta$. Each term $r^\nu \sin\nu\theta$ is harmonic by Step 1 of (b).

Step 4. Cartesian form

$r\sin\theta = y$, and $r^3 \sin 3\theta = r^3(3\sin\theta\cos^2\theta - \sin^3\theta) = 3(r\cos\theta)^2(r\sin\theta) - (r\sin\theta)^3 = 3x^2y - y^3$ (using $\sin 3\theta = 3\sin\theta - 4\sin^3\theta = 3\sin\theta\cos^2\theta - \sin^3\theta$). Therefore

$$u(x,y) = \frac{3y}{4a} - \frac{3x^2y - y^3}{4a^3}.$$

Direct check: $u_{xx} = -\frac{6y}{4a^3}$, $u_{yy} = \frac{6y}{4a^3}$, so $\Delta u = 0$. On the insulated edge $x = 0$ ($\theta = \pi/2$), $u_x = -\frac{6xy}{4a^3} = 0$: the normal derivative vanishes, as required. On $y = 0$ ($\theta = 0$), $u = 0$.

Step 5. Values

- $(r,\theta) = (a/2, \pi/2)$: $\sin\frac{\pi}{2} = 1$, $\sin\frac{3\pi}{2} = -1$, so $u = \frac{3}{4}\cdot\frac{1}{2} + \frac{1}{4}\cdot\frac{1}{8} = \frac{3}{8} + \frac{1}{32} = \frac{13}{32} \approx 0.406$.
- $(r,\theta) = (a/2, \pi/6)$: $\sin\frac{\pi}{6} = \frac{1}{2}$, $\sin\frac{\pi}{2} = 1$, so $u = \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{1}{2} - \frac{1}{4}\cdot\frac{1}{8} = \frac{3}{16} - \frac{1}{32} = \frac{5}{32} \approx 0.156$.

Both lie between the minimum 0 and the maximum 1 of the boundary values, as the maximum principle for Laplace's equation requires (Haberman Sec. 2.5).

KEY RESULTS

- $\nu_n = 2n - 1$; $\sin^3\theta = \frac{3}{4}\sin\theta - \frac{1}{4}\sin 3\theta$; $A_1 = \frac{3}{4}$, $A_2 = -\frac{1}{4}$, $A_n = 0$ for $n \geq 3$.
- $u(r,\theta) = \frac{3}{4}\frac{r}{a}\sin\theta - \frac{1}{4}\left(\frac{r}{a}\right)^3 \sin 3\theta$; Cartesian: $u = \frac{3y}{4a} - \frac{3x^2y - y^3}{4a^3}$.
- $u(a/2, \pi/2) = \frac{13}{32}$, $u(a/2, \pi/6) = \frac{5}{32}$.

Interpretation and checks

Phase 4. Asymptotic and physical checks

- **Limits:** $r \to 0$: $u \approx \frac{3}{4}\frac{r}{a}\sin\theta \to 0$, the vertex takes the temperature of the cold edge; the leading mode r^{ν_1} dominates near the vertex. For a general opening, $\nu_1 = \pi/(2\alpha) < 1$ when $\alpha > \pi/2$: u stays bounded but $|\nabla u| \sim r^{\nu_1 - 1} \to \infty$ at the vertex (a corner singularity caused by the mixed conditions; with $u = 0$ on both edges it would only appear for $\alpha > \pi$). The quarter disc, $\nu_1 = 1$, is exactly the borderline case with a finite gradient. Along the insulated edge, $u(r, \pi/2) = \frac{3}{4}s + \frac{1}{4}s^3$ with $s = r/a$ increases monotonically from 0 to 1.
- **Checks:** the PDE, the two edge conditions, boundedness and the arc datum are satisfied (script, in polar and Cartesian form); the coefficient formula of (b) reproduces $A_1 = \frac{3}{4}$, $A_2 = -\frac{1}{4}$ and $A_n = 0$ for $n = 3, 4, 5$; the norm $\int_0^{\pi/2} \sin^2 3\theta \, d\theta = \frac{\pi}{4} = \frac{\alpha}{2}$. Dimensionless problem: $(r/a)^\nu$ is a pure number for any a.
- **Common mistakes in the exam:** (1) using the full-disc eigenfunctions $\cos n\theta$, $\sin n\theta$ or the half-disc ones $\sin n\theta$ ($\nu = n$): the insulated edge requires odd quarter-wave multiples; (2) keeping $r^{-\nu}$ or a $\ln r$ term, which blow up at the vertex; (3) computing the coefficients by integrals over $(0, \pi)$ or with norm $\pi/2$ instead of $\alpha/2 = \pi/4$; (4) not recognising $\sin^3\theta$ as a finite sum and writing an infinite series with wrong integrals.